

LESSON 5

Valid Arguments & Fallacies

Name: _____ Date: _____

Arguments: premises and conclusion

An **argument** is not a fight — it is a structured list:

- **Premises** — statements you assume for the moment.
- **Conclusion** — what you claim follows from them.

EXAMPLE ARGUMENT

1. All mammals breathe air. (premise)
2. Whales are mammals. (premise)
3. Therefore whales breathe air. (conclusion)

Valid vs sound

- **Valid:** IF the premises were true, the conclusion *must* be true. Structure is correct.
- **Sound:** Valid **and** every premise is actually true in the real world.

A valid argument can still have a false conclusion if a premise is false. Validity is about the *pattern*, not whether you agree with the facts.

Valid patterns to recognize

MODUS PONENS

$p \rightarrow q \cdot p \cdot \text{therefore } q$

If it rains, the field is muddy. It rains. $\therefore$ The field is muddy. $\checkmark$

MODUS TOLLENS

$p \rightarrow q \cdot \neg q \cdot \text{therefore } \neg p$

If the alarm is armed, the light is red. The light is not red. $\therefore$ The alarm is not armed. $\checkmark$

Invalid patterns — classic traps

AFFIRMING THE CONSEQUENT $\times$

From $p \rightarrow q$ and q , concluding p .

"If you study hard, you pass. You passed. So you studied hard." — Maybe, but not forced by logic.

DENYING THE ANTECEDENT $\times$

From $p \rightarrow q$ and $\neg p$, concluding $\neg q$.

"If it rains, the game is canceled. It didn't rain. So the game wasn't canceled." — Wrong; they might cancel for wind.

More fallacies (everyday names)

- **False dilemma:** "Either you're with us or against us" when middle options exist.
- **Hasty generalization:** "My friend got sick after sushi, so all sushi is unsafe."
- **Ad hominem:** Attacking the person instead of the argument.

De Morgan's laws (simplifying negations)

- $\neg(p \wedge q) \equiv (\neg p \vee \neg q)$ — "Not both" = "at least one is false."
- $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$ — "Neither" = "not this and not that."

WORKED EXAMPLE

"It is not true that the store is open and I have cash."

Let p = store open, q = I have cash. Original: $\neg(p \wedge q)$. De Morgan: $\neg p \vee \neg q$ = "The store is closed OR I don't have cash (or both)."

Practice — Lesson 5

5.1 WARM-UP

Label **premises** and **conclusion**:

“Soccer is a sport. All sports require practice. Therefore soccer requires practice.”

P1, P2, CONCLUSION

5.2 WARM-UP

Rewrite using De Morgan: $\neg(p \vee q)$. Then put it in plain English if p = “I like math” and q = “I like art.”

SYMBOLIC + ENGLISH

5.3 STANDARD

Valid or invalid? Name the pattern if you can.

1. If n is divisible by 10, then n is divisible by 5. n is divisible by 10. So n is divisible by 5.
2. If you wear a helmet, you are safe. You are safe. So you wore a helmet.
3. If it is Friday, the cafeteria serves pizza. It is not Friday. So no pizza today.

VALID/INVALID + PATTERN NAME

5.4 STANDARD

Fix this invalid argument by changing **only the conclusion** so the argument becomes valid (premises stay the same):

P1: If a bird is a penguin, it cannot fly.

P2: Tweety is not a penguin.

Bad conclusion: Tweety can fly.

NEW VALID CONCLUSION + WHY

5.5 STANDARD

Build a truth table for $(p \rightarrow q) \wedge p$. When the last column is true, what must be true about q ?
Connect to modus ponens.

TABLE + CONNECTION

5.6 CHALLENGE

Write a **valid** argument with two premises and one conclusion about a topic you know (sport, game, school rule). Then write a **invalid** argument on the same topic that looks similar but commits affirming the consequent or denying the antecedent.

VALID ARGUMENT / INVALID ARGUMENT

5.7 CHALLENGE

Paraphrase this ad-style claim and identify any fallacy: “Either you buy our headphones or you don’t care about music quality.” Rewrite it as an honest logical statement that does not trap the listener.

FALLACY NAME + HONEST REWRITE